

1 Question (15 points)

Describe a multitape Turing machine with $k + 1$ tapes that recognizes the language $L = \{x_1^1 x_2^2 \dots x_k^k \mid k \geq 1\}$, $\Sigma = \{x_1, x_2, \dots, x_k\}$. You need to use all tapes in your solution. Do not design the TM. Describe the steps it needs to take.

2 Question (15 points)

Design a TM that recognizes the language $L = \{a^n b^{nm} | n, m \geq 1\}$. Do not describe the steps of TM. Draw a state diagram, give states, transitions, etc.

3 Question (15 points)

If there is a state in a DFA that is not reachable from the start state, it is redundant. Consider the language:

$$E_{DFA} = \{ \langle D \rangle \mid D \text{ is DFA and } D \text{ does not contain redundant state} \}$$

Is E_{DFA} decidable? Prove it.

4 Question (15 points)

Describe a TM that recognizes the language $L = \{a^n b^m c^{n-m} \mid n, m \geq 1, n \geq m\}$. Do not describe the steps of TM. Draw a state diagram, give states, transitions, etc.